

Homework 5

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Problem 1.

(a) Yes.

Let $x, y \in S$,

where $x = x_1 + x_2, y = y_1 + y_2$,

$x_1, y_1 \in S_1, x_2, y_2 \in S_2$

So we have:

$$\begin{aligned}\lambda x + (1 - \lambda)y &= \lambda(x_1 + x_2) + (1 - \lambda)(y_1 + y_2) \\ &= \lambda x_1 + (1 - \lambda)y_1 + \lambda x_2 + (1 - \lambda)y_2\end{aligned}$$

Since S_1, S_2 are convex sets, we have:

$$\lambda x_1 + (1 - \lambda)y_1 \in S_1,$$

$$\lambda x_2 + (1 - \lambda)y_2 \in S_2$$

Thus, we have:

$$\lambda x + (1 - \lambda)y \in S$$

Therefore, S is a convex set.

(b) Yes.

$$\begin{aligned}ax^2 + 2x + \frac{1}{a} &= \left(\sqrt{a}x + \frac{1}{\sqrt{a}}\right)^2 \geq 0 \\ &\Rightarrow x \in \mathbb{R}\end{aligned}$$

Thus, $\{x \in \mathbb{R} : ax^2 + 2x + \frac{1}{a} \geq 0\}$ is a convex set.

(c) Yes.

$$\begin{aligned}\lambda y_1 + (1 - \lambda)y_2 &\geq \lambda f(x_1) + (1 - \lambda)f(x_2) \\ &\geq f(\lambda x_1 + (1 - \lambda)x_2)\end{aligned}$$

Thus, $\{(y, x) \in \mathbb{R} \times \mathbb{R}^n : y \geq f(x)\}$ is a convex set.

(d) No. Let $f(x) = x^2, y = x^2$, so point $(1, 1)$ and $(-1, 1)$ are in the set, but point $(0, 2)$ is not in the set.

Thus, the set $\{(x, y) \in \mathbb{R}^n : y = f(x)\}$ is not a convex set.

(e) No. Hessian matrix: $H = \begin{bmatrix} 4 & -2 & 8 \\ -2 & 6 & 4 \\ 8 & 4 & -10 \end{bmatrix}$ is not positive semidefinite.

Thus, the set $\{x \in \mathbb{R}^3 : x^T A x \leq 0\}$ is not a convex set.

Problem 2.

(a) Yes.

$$\begin{aligned}\lambda f(x) + (1 - \lambda)f(y) &= \lambda \|x\|_1 + (1 - \lambda)\|y\|_1 \\ &\geq \|\lambda x + (1 - \lambda)y\|_1 \\ &= f(\lambda x + (1 - \lambda)y)\end{aligned}$$

Thus, $f(x) = \|x\|_1$ is a convex function.

(b) Yes.

$$\begin{aligned}\lambda f(x) + (1 - \lambda)f(y) &= \sum_{i=1}^d [\lambda f_i(x_i) + (1 - \lambda)f_i(y_i)] \\ &\geq \sum_{i=1}^d f_i(\lambda x_i + (1 - \lambda)y_i) \\ &= f(\lambda x + (1 - \lambda)y)\end{aligned}$$

Thus, f is a convex function.

(c) Yes.

$$\begin{aligned}\lambda f(x) + (1 - \lambda)f(y) &= f(\lambda x) + f((1 - \lambda)y) \\ &\geq f(\lambda x + (1 - \lambda)y)\end{aligned}$$

Thus, f is a convex function.

(d) Yes.

$$\begin{aligned}\lambda f(x) + (1 - \lambda)f(y) &= \lambda|x|^{\frac{3}{2}} + (1 - \lambda)|y|^{\frac{3}{2}} \\ &\geq |\lambda x + (1 - \lambda)y|^{\frac{3}{2}} \\ &= f(\lambda x + (1 - \lambda)y)\end{aligned}$$

Thus, $f(x) = |x|^{\frac{3}{2}}$ is a convex function.

(e) Yes.

$$f'(x) = \frac{1}{\sqrt{1+x^2}} + 2x, \quad f''(x) = -\frac{x}{(1+x^2)^{\frac{3}{2}}} + 2 \geq 0$$

Thus, $f(x) = \ln(x + \sqrt{1+x^2}) + x^2$ is a convex function.

Problem 3.

(a)

$$\begin{aligned}\frac{\partial f}{\partial \beta_0} &= 2 \sum_{i=1}^n (\beta_0 + \beta^T x_i - y_i) = 2n\beta_0 \\ \frac{\partial f}{\partial \beta} &= 2 \sum_{i=1}^n (\beta_0 + \beta^T x_i - y_i) x_i + 2\lambda \beta = 2\beta_0 \sum_{i=1}^n x_i\end{aligned}\tag{1}$$

Let $\frac{\partial f}{\partial \beta_0} = 0, \frac{\partial f}{\partial \beta} = 0$, we have: $\beta_0^* = 0$

(b)

$$\begin{aligned}\tilde{f}(\beta_0 - \beta^T q + r, \beta) &= \sum_{i=1}^n [\beta_0 - \beta^T q + r + \beta^T (x_i + q) - y_i]^2 \\ &= \sum_{i=1}^n (\beta_0 + \beta^T x_i - y_i + r)^2 \\ &= f(\beta_0, \beta)\end{aligned}$$

Problem 4.

1. Let $f = 3x_1^2 + 4x_2^2 + 2x_1x_2$, we have:

$$\begin{aligned} f &= 3x_1^2 + 4x_2^2 + 2x_1x_2 \\ \nabla f &= \begin{bmatrix} 6x_1 + 2x_2 \\ 2x_1 + 8x_2 \end{bmatrix} \\ \nabla^2 f &= \begin{bmatrix} 6 & 2 \\ 2 & 8 \end{bmatrix} \end{aligned}$$

Since $\nabla^2 f$ is positive definite, f is a convex function.

Let $g = \sqrt{1 + x_1^2 + x_2^2} + x_1 - 2$, we have: Since $\sqrt{1 + x_1^2 + x_2^2}$ is a convex function and $x_1 - 2$ is a linear function, g is a convex function.

Thus, the optimization problem is a convex optimization problem.

2. KKT conditions:

$$\nabla f + \lambda \nabla g = 0$$

$$\lambda g = 0$$

$$\lambda \geq 0, g \leq 0$$

3. (i) When $\lambda = 0$, we have:

$$\begin{aligned} \nabla f + \lambda \nabla g &= 0 \\ \Rightarrow \begin{bmatrix} 6x_1 + 2x_2 \\ 2x_1 + 8x_2 \end{bmatrix} &= 0 \\ \Rightarrow (x_1, x_2) &= (0, 0) \end{aligned}$$

Since it is a convex optimization problem, $(x_1, x_2) = (0, 0)$ is the global minimum.

$(0, 0)$ is the only KKT point.

Problem 5.

1. The solution and number of iterations are:

Method	Approximate Solution	Iterations
Bisection Method	0.6931	14
Golden Section Method	0.6931	20